Trainings

Select Service Tools

Recommended Cummins® Service Tools

- Dial indicator, Part Number 3824564
- Overheard adjustment kit, Part Number 5298979

Additional Service Items

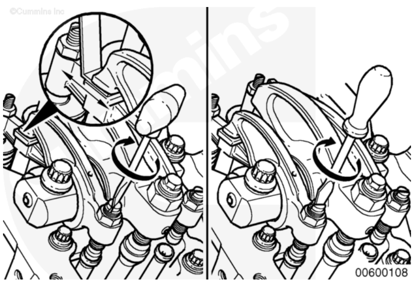
- Feeler gauge
- Magnetic block with extension arm

General Information

Cummins Inc. has found that engines in most applications will **not** experience significant valve/injector train wear. It is recommended that the valve and injector adjustment be performed **only** when the injectors are removed, when other repairs disturb the valve train, or, when the valves or injectors are checked and found out of adjustment.

Valves and injectors **must** be correctly adjusted for the engine to operate efficiently. Valves and injector adjustment **must** be performed using the values listed in this section.

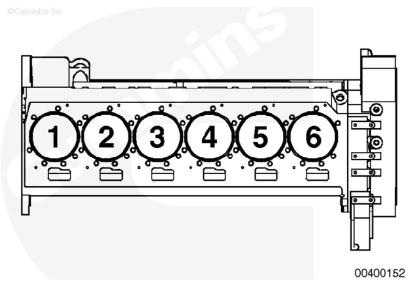
Injector and Valve overhead **must** be checked for clearance at specified maintenance interval. Use the following section in the QSK23 Operation and Maintenance Manual, Bulletin 4021374. Refer to Section 6.



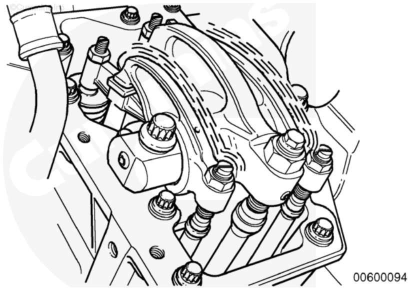
The cylinders are numbered from the front of the engine.

Firing order for the QSK23 engine is 1-5-3-6-2-4.

The crankshaft rotation is **clockwise** when viewed from the front of the engine.



Each cylinder has three rocker levers. When facing the cylinder head from the intake side of the engine, the lever on the left is the intake rocker lever, and the lever on the right is the exhaust rocker lever. The center rocker lever is for the injector.

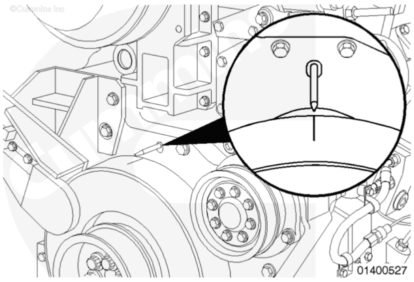


The engine has valve and injector adjustment marks on the vibration damper.

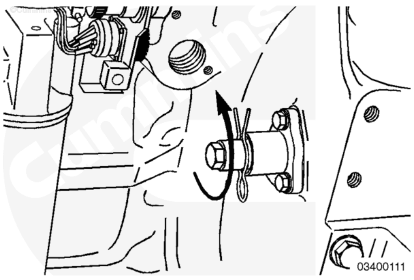
Valve and injector marks **must** be aligned with the pointer, or a false adjustment can occur.

One pair of valves and one injector are adjusted at each vibration damper index mark.

Two crankshaft revolutions are required to adjust all of the valves and injectors.

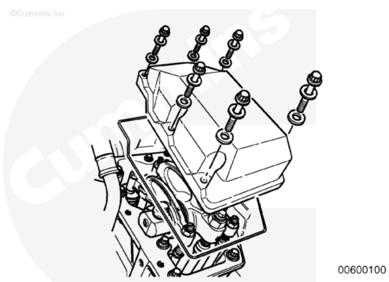


This illustration shows the engine barring device. To use the device, remove the clip and push the device towards the flywheel. The barring device **must** be rotated **counterclockwise** to turn the flywheel and crankshaft in the direction of normal rotation.



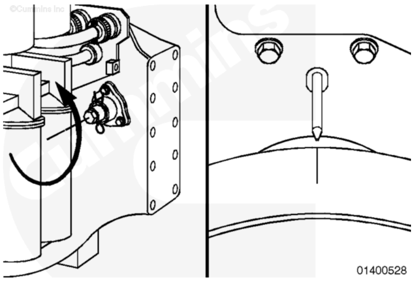
Preparatory Steps

- Remove the rocker lever covers. Refer to Procedure 003-011 (Rocker Lever Cover) in Section 3. (/qs3/pubsys2/xml/en/procedures/89/89-003-011.html)



Measure

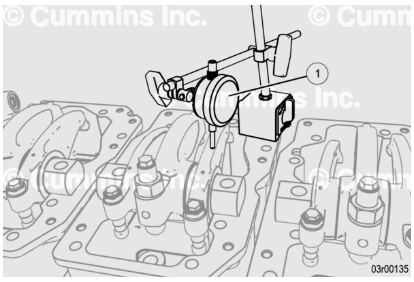
Remove the retaining clip on the barring device and using the barring device, rotate the crankshaft in the direction of normal rotation until the pointer on the front of the engine aligns with the 1 to 6 top line on the crankshaft pulley.



Refer to vibration damper mark determination chart in this procedure.

Cylinder number one should now be at top dead center compression. With the piston in this position both valves are closed and the valve rocker arms for cylinder number one should be loose. Observe the injector pushrods for cylinders two and five. Cylinder two injector pushrod should be higher and the injector plunger bottomed in the cup. If this is the case, the valves on cylinder one can be checked and the injector lift is done on cylinder two. If the valves are **not** loose on cylinder number one, check the valves on cylinder six. Those valves should be loose and number five injector pushrod should be higher than two. If this is true check the injector lift on cylinder five and the valves on number six.

Injector Lift:



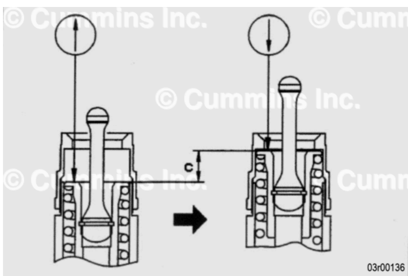
To determine which cylinder is ready for checking, identify the cylinders that would be in position for injector lift checking, refer to the Valve and Injector Adjustment Determination chart in this procedure. For example, if the engine was barred to the "1.6 TOP" mark, cylinder 2 or 5 would be in position for injector plunger lift check.

To check injector plunger lift the engine cylinder 1 and 6 **must** be at Top Dead Center (TDC).

Set dial gauge (1) to the head of the plunger to be checked.

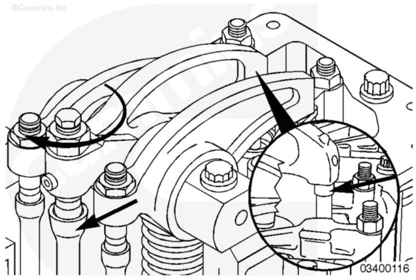
Use dial gauge with a stroke of at least 30 mm or 1.181 in.

Crank the crankshaft and measure lift (see letter "c") of the plunger at the point where deflection on the dial gauge is maximum. If measurement is **not** within specified value, go to Adjust section in this procedure.



If injector lift "c" is within specified value, proceed to Valve Check.

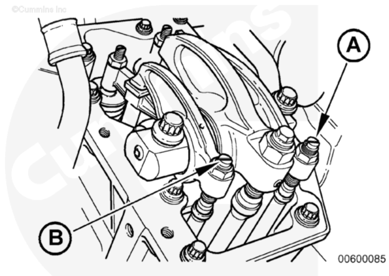
Valve Lash:



To determine which cylinder valve is ready for checking, identify the cylinders that would be in position for valve checking, refer to the Valve and Injector Adjustment Determination chart in this procedure. For example, if the engine was barred to the “1.6 TOP” mark, cylinder 1 or 6 would be in position for intake and exhaust valve clearance check.

The engine **must** be cold to perform the overhead set checking procedure.

Select a feeler gauge for the correct valve lash specification.



Valve Clearance - Check			
	mm		in
Exhaust valves (A)	0.60	MIN	0.024
	0.64	MAX	0.025
Intake valves (B)	0.30	MIN	0.012
	0.34	MAX	0.013

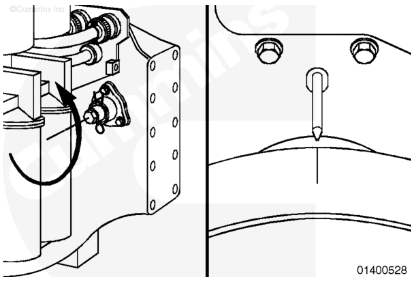
Insert the feeler gauge between the top of the crosshead and the rocker lever nose pad. Verify the feeler gauge is completely under the pivoting rocker lever nose pad.

Continue Checking Injector Lift and Valve Clearance:

Bar the engine to next damper timing mark and (refer to the chart in this procedure) to the determine the next injector cylinder and valve cylinder to check . If the rocker lever assemblies were **not** removed, identify the cylinders that are in position for injector plunger lift check. On these two cylinders, identify the injector push rod which is higher in relation to the top of the rocker lever housing. This is the cylinder ready for injector lift checking. Refer to the chart in this section to determine the corresponding valve clearance check cylinder. Repeats steps above until all cylinders are checked for injector lift and valve clearance.

Once all injector plunger lift and valve clearance checks have been found to be within specification in this procedure, proceed to Finishing Steps section. If injector plunger or valve clearance does **not** meet specification, proceed to the Adjust section for adjusting the overhead.

Adjust

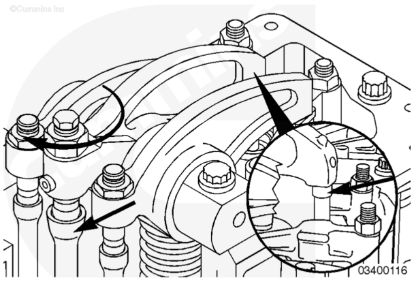


If the rocker lever assemblies were removed for this repair, use this step to determine the correct cylinder to set.

Lubricate the adjusting screw threads with clean engine oil prior to making valve and injector adjustments.

All adjusting screws **must** be loose on all cylinders and the push rods **must** remain in alignment.

Bar the engine to the next valve set mark on the damper. Setting can begin on any valve set mark.

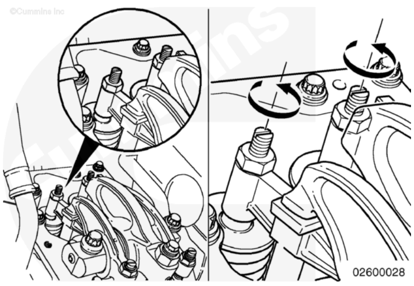


To determine which cylinder is ready for setting, identify the cylinders that would be in position for injector setting, refer to the chart in this procedure. For example, if the engine was barred to the “1.6 TOP” mark, cylinder 2 or 5 would be in position for injector setting.

For these two cylinders, turn the injector adjusting screws and locknuts down until the injector rocker is in contact with the push rod and injector link. Perform the injector setting on the cylinder which has the most visible threads above the locknut on the injector adjusting screw.

Prior to barring the engine, the injector which had less visible threads and was **not** set in the previous step **must** have the injector adjusting screw backed out until there are at least as many threads above the adjusting nut as for the injector that was set.

To determine which cylinder is ready for valve setting, refer to the chart in this section and identify the cylinder which corresponds to the injector set cylinder from the previous step. For example, if cylinder 2 injector was set, then cylinder 1 would be ready for valve setting.



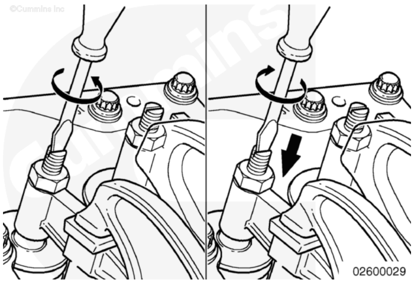
Note : The engine must be cold to perform the overhead set procedure.

Crosshead adjustment **must always** be made before attempting to adjust the valves.

Adjust the crossheads on the cylinder that is ready for valve setting.

Loosen the crosshead adjusting screw locknuts on the intake and exhaust valve crossheads.

Use the following procedure to adjust both the intake and exhaust crossheads.

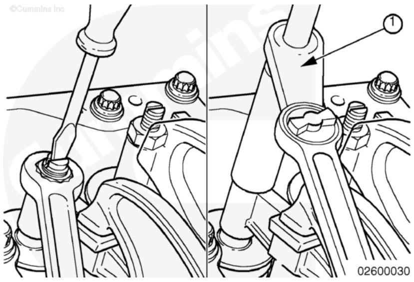


Turn the adjusting screw **counterclockwise** at least one turn.

Hold the crosshead down against the guide.

Turn the adjusting screw **clockwise** until it touches the top of the valve stem, but does **not** raise the crosshead.

Hold the adjusting screw in this position. The adjusting screw **must not** turn when the locknut is tightened to its torque value.



Tighten the locknut.

The following torque values are given with and without a torque wrench adapter, Part Number 3163196.

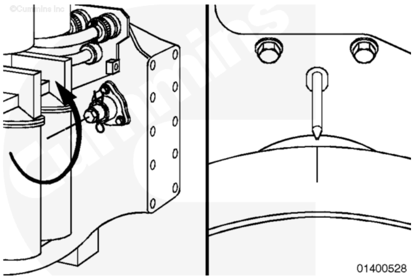
Torque Value:
With adapter

1. 60 n•m [44 ft-lb]

Torque Value:
Without adapter

1. 65 n•m [48 ft-lb]

Bar the engine to the next damper timing mark and (refer to the chart in this procedure) to determine the next injector cylinder and valve cylinder to set.



If the rocker lever assemblies were **not** removed, identify the cylinders that are in position for injector setting.

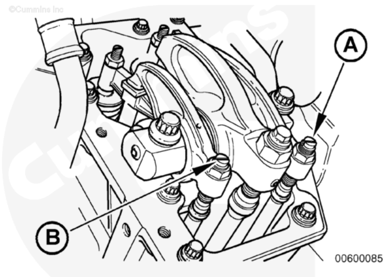
On these two cylinders, identify the injector push rod which is higher in relation to the top of the rocker lever housing. This is the cylinder ready for injector setting.

Refer to the chart in this section to determine the corresponding valve set cylinder.

Valve and Injector Adjustment Determination		
Vibration Damper Mark	**Valve Adjustment on Cylinder Number**	**Injector Adjustment on Cylinder Number**
1.6 TOP	1	2

Valve and Injector Adjustment Determination		
Vibration Damper Mark	**Valve Adjustment on Cylinder Number**	**Injector Adjustment on Cylinder Number**
2.5 TOP	5	4
3.4 TOP	3	1
1.6 TOP	6	5
2.5 TOP	2	3
3.4 TOP	4	6

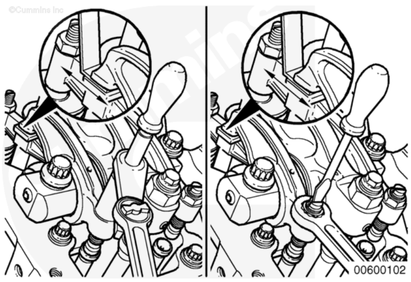
Valve Adjustment



Valve Clearances - Initial Set			
	mm		**in**
Exhaust valves (A)	0.62	MAX	0.024
Intake valves (B)	0.32	MAX	0.013

Valve Clearances - Check			
	mm		**in**
Exhaust valves (A)	0.60	MIN	0.023
	0.64	MAX	0.025
Intake valves (B)	0.30	MIN	0.012
	0.34	MAX	0.013

The Torque Wrench Method and Screwdriver Method are used to set valve lash clearance. Both are described below. Either method can be used; however, the Torque Wrench Method has proven to be the most consistent.



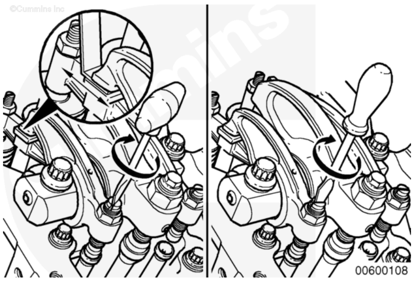
Make sure the crossheads have been adjusted and are firmly in place on the valve stems.

Make sure the feeler gauge is under the center of the rocker lever or incorrect adjustment can result.

The adjustment screws **must** turn freely or a false reading or setting can occur.

Select the proper feeler gauge for the valves being set. Use feeler gauge set, Part Number 3823557, or equivalent.

Valve Adjustment - Torque Wrench Method



Make sure parts are aligned and squeeze the oil out of the valve train by tightening the adjusting screw.

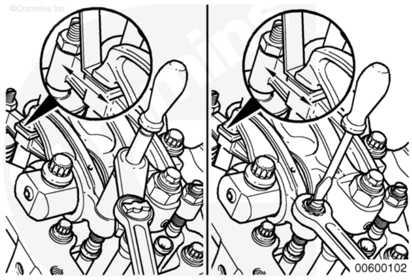
Loosen the adjusting screw at least one revolution.

Insert the feeler gauge between the rocker lever and the crosshead.

Use torque wrench, Part Number 3376592, to tighten the adjusting screw.

Torque Value: 0.68 n•m [6 in-lb]

Remove the feeler gauge.



The adjusting screw **must not** turn when the locknut is tightened. Locknut torque can be applied with or without the torque wrench adapter, Part Number 3163196.

Tighten the locknut.

Torque Value:
With adapter

1. 48 n•m [35 ft-lb]

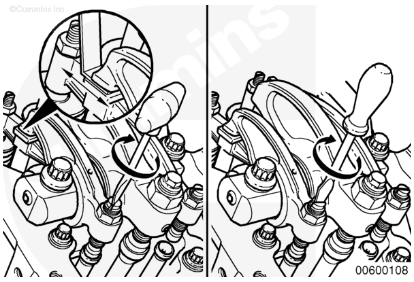
Torque Value:
Without adapter

1. 68 n•m [50 ft-lb]

Attempt to insert a feeler gauge that is 0.03 mm [0.001 in] thicker. The valve lash is **not** correct if the thicker feeler gauge will fit.

Repeat the adjustment process until the proper lash is obtained.

Valve Adjustment - Screwdriver Method



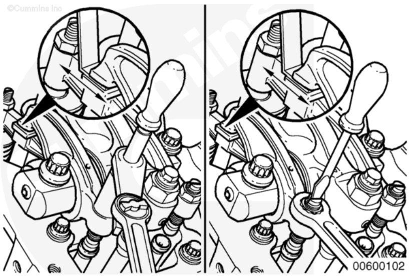
Make sure the parts are aligned and squeeze the oil out of the valve train by tightening the adjusting screw.

Loosen the adjusting screw at least one revolution.

Insert the feeler gauge between the rocker lever and the crosshead.

Tighten the adjusting screw until the rocker lever touches the feeler gauge.

The adjusting screw **must not** turn when the locknut is tightened. Locknut torque can be applied with or without a torque wrench adapter, Part Number 3163196.



Tighten the locknut.

Torque Value:
With Adapter

1. 48 n•m [35 ft-lb]

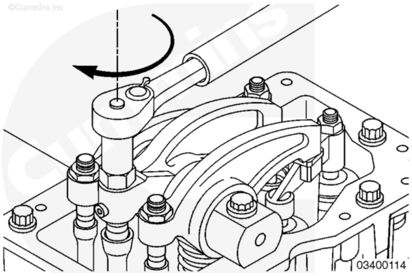
Torque Value:
Without Adapter

1. 68 n•m [50 ft-lb]

Attempt to insert a feeler gauge that is 0.03 mm [0.001 in] thicker. The valve lash is **not** correct if the thicker feeler gauge will fit.

Repeat the adjustment process until the proper lash is obtained.

Injector Lever Adjustment



Either a click type or a dial type torque wrench can be used to tighten the injector rocker lever adjusting screw. The specified adjusting screw torque **must** fall near the center of the operating range for the torque wrench used. If the screw chatters during setting, repair the screw and lever as required.

Tighten the adjusting screw.

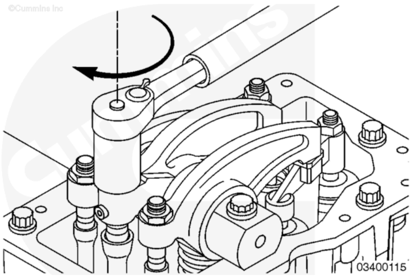
Torque Value: 32 n•m [24 ft-lb]

Loosen the adjusting screw at least one revolution.

Tighten the adjusting screw again.

Torque Value: 32 n•m [24 ft-lb]

When tightening the adjusting screw locknut, do **not** hold the adjusting screw in position.

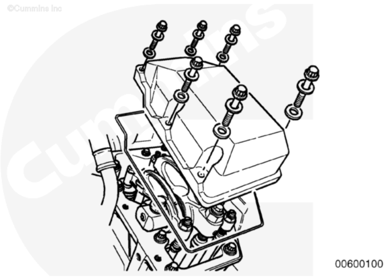


Tighten the injector adjusting screw locknut.

Torque Value: 225 n•m [166 ft-lb]

Finishing Steps

- Install the rocker lever covers. Refer to Procedure 003-011 (Rocker Lever Cover) in Section 3. (/qs3/pubsys2/xml/en/procedures/89/89-003-011.html)
- Operate the engine and check for leaks.



Last Modified: 07-Jun-2024